

Aarón Ayllón Benítez, PhD

AI & Data Science Leader — Enterprise ML, GenAI & Agentic Platforms • Team and Portfolio Ownership

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EXECUTIVE SUMMARY

Data scientist turned AI/ML leader (PhD Computer Science, MSc Artificial Intelligence) who has spent five-plus years building and running multidisciplinary data science teams inside a global corporation. Scaled a team from 8 to 16 data scientists, ML/software engineers and solution designers, and owned two production AI/ML platforms covering the full analytics journey from descriptive to predictive, delivering recurring savings above €3M. Owns the portfolio calls that follow — what to scale, what to hand over and what to shut down — alongside demand intake and prioritisation across departments and European markets. Combines hands-on technical depth in ML, LLMs and agentic workflows with life-sciences training in biochemistry, bioinformatics and biomedical machine learning on patient-derived data, and a track record of translating analytics into quantified business cases and defending them in front of senior leadership.

€3M+	8 → 16	~90%	~100
recurring savings from portfolio decisions	team scaled, hired and developed	less effort per new project after standardisation	the whole of BASF's global data science community

CORE SKILLS

AI & Data Science: Machine Learning, Deep Learning, Computer Vision, Time-series Forecasting, Descriptive → Predictive Analytics, GenAI & LLMs, Agentic workflows (multi-step reasoning, tool calling, evaluation), Knowledge Graphs & Ontologies, MLOps

Leadership: Hiring & team building, Mentoring managers and junior scientists, Performance and underperformance management, Career paths, Executive communication & storytelling, Cross-functional alignment (business + tech + data), Multi-market coordination

Portfolio & Delivery: Portfolio governance & rationalisation, Product line management, Build / scale / hand-over / sunset decisions, Product strategy & roadmap, Demand intake & prioritisation, Enterprise platform delivery, Change management

Technical: Python, R, SQL, advanced statistics, Azure & Azure DevOps, APIs / MCP, data engineering workflows, CI/CD, Power BI, AI development tools

Domains: Life sciences, healthcare & biomedical data, pharma and agricultural R&D, industrial and commodity business

PROFESSIONAL EXPERIENCE

BASF Digital Solutions S.L. — Madrid, Spain

Head of Advanced Imaging & Forecasting Solutions — AI & ML Platforms

2022 – Present

Official BASF title: Product Manager • scope equivalent to Head of Product / Global Product Line Manager

- Lead a team of 16 (data scientists, ML and software engineers, solution designers) scaled from 8, owning hiring, career paths, mentoring of both managers and junior scientists, performance and underperformance management.
- Own portfolio strategy, investment priorities and product roadmaps across computer vision, MLOps, forecasting and semantic infrastructure, serving BASF's global data science community.
- Led portfolio decisions with direct financial impact: transferred the computer vision / MLOps platform to its best-fit owner (~€2M recurring savings, 70 reusable templates, ~90% effort reduction for ~100 data scientists) and recommended closure of the forecasting platform after an ROI and adoption review (~€1.5M savings).
- Deliver the full analytical journey from descriptive to predictive: a self-service forecasting platform giving business users time-series analytics and decision support, and a computer vision platform for classification, segmentation and object detection — both deployed at scale in the cloud.
- Own demand intake, prioritisation and delivery for business stakeholders across three departments and multiple European markets, running biweekly strategic reviews to align roadmaps, arbitrate competing priorities and resolve conflicts.
- Secured €200K in internal funding for GenAI-powered intelligent labelling and automation of image-analysis workflows; drove an AI-native engineering culture with LLM integration into ML pipelines and team-wide adoption of AI development tools.
- Won backing for five innovation initiatives worth €360K by turning analytical opportunities into quantified business cases and presenting them to senior leadership with clear data storytelling.
- Acted as Business Transformation Coordinator for Spain, Portugal and Italy through a legally constrained restructuring, aligning business units and site managing directors while the department was integrated into the commodity business.
- Partnered as a technical peer with senior ML engineers on architecture, model choices and deployment trade-offs, keeping decision quality high and communication with leadership quantitative and concrete.

BASF Digital Solutions S.L. — Madrid, Spain

Technical Product Manager — Ontology & Semantic Layer

2021 – 2022

Official BASF title: Product Component Manager • foundation for the AI & ML platform portfolio taken on next

- Owned enterprise ontology and knowledge-graph strategy — the semantic foundation that makes heterogeneous data interoperable and machine-readable for AI and agentic systems.
- Led 5 engineers and 2 external partners to deliver 20 bio-ontologies adopted as the vocabulary standard across the R&D Agricultural Solutions department by every internal tool within 6–12 months.
- Co-authored BASF's GOMO ontology governance framework (published, ISMB 2022) with the Ontology Engineering Group at Universidad Politécnica de Madrid.
- Led the EPPO ontology for plant and pest code representation, published in Frontiers in Artificial Intelligence and released as open source.
- Established and led a biweekly data steward community of 10 governing vocabulary standardisation, processing 10–15 terminology requests per month.

Umeå Plant Science Centre — Umeå, Sweden

Postdoctoral Researcher, Machine Learning

2020 – 2021

- Kempestiftelserna Fellowship: applied machine learning and gene regulatory network inference to improve functional annotation at genome scale.
- Designed production-oriented ML pipelines with reproducibility, auto-scaling and experiment tracking, translating research workflows into deployable systems.
- Secured 2 competitive research grants as main author (KSLA, SPPS) and served as Guest Editor for a Q1 journal special issue.

Université de Bordeaux / LaBRI — Bordeaux, France

PhD Researcher, Computer Science

2016 – 2019

- Ministerial funding (MESR). Built GSA_n, a gene-set annotation web platform still live and in use (gsan.labri.fr, 18+ citations), and ontology-based ML methods for semantic similarity and knowledge representation.
- Published across Q1 journals and IEEE; co-supervised a Master's student with the thesis director; first prize, Scientific Game Jam Bordeaux.
- Ran an externally funded applied-ML engagement alongside the thesis: MHC epitope prediction in HIV patients from patient-derived sequence data — hands-on biomedical machine learning in a clinical research setting.

INSERM — Bordeaux, France

Bioinformatics Engineer

2015 – 2016

- Developed algorithms for multi-omics data integration and enrichment analysis in a clinical research setting, under Rodolphe Thiébaud — later a co-author on the resulting doctoral publications.

LEADERSHIP APPROACH

- **Build the bench, not just the backlog.** Doubled the team through deliberate hiring, individual career paths and coaching of the managers who report into the role — including the hard conversations around underperformance rather than routing around them.
- **Decide with evidence, and be willing to kill your own product.** Recommended closing a platform I owned once the ROI and adoption data no longer supported it, and transferred another to a better-fit owner — both decisions defended with numbers in front of senior leadership.
- **Translate between worlds.** Comfortable challenging model choices with ML engineers in the morning and framing the same problem as a business case for executives in the afternoon; this is where most analytics value is won or lost.

EDUCATION

MSc Artificial Intelligence — Universidad Internacional de La Rioja (UNIR)

2023–2024

PhD Computer Science — Université de Bordeaux / LaBRI, ministerial funding (MESR)

2016–2019

MSc Bioinformatics — University of Murcia

2013–2015

BSc Biochemistry — University of Murcia

2009–2013

RESEARCH & INDUSTRY VISIBILITY

- **Selected publications:** Nature Communications (2021, contributor); NAR Genomics and Bioinformatics (Q1, 2020, first author); PLoS ONE (2018, first author); IEEE Information Visualization, London (2017, first author, invited talk); EPPO Ontology, Frontiers in Artificial Intelligence (2023, first author); GOMO ontology governance model (ISMB 2022). h-index 5; 132 citations.
- **Speaking & media:** Industry showcase speaker at EXPOQUIMIA 2023 on advanced image analysis for the chemical industry, plus roundtable participation; podcast guest on AI in industry (Tecnología, Pasión y Futuro) and with the AI Cluster of the Community of Madrid.
- **Editorial & funding:** Guest Editor, Plants (MDPI, Q1) special issue on gene annotation; Kempestiftelserna Fellowship and 2 competitive grants secured as main author (KSLA, SPPS).
- **Teaching & mentoring:** 100+ hours of university teaching (Umeå, Bordeaux); mentored 25+ Master's and Bachelor's students across four academic years.

ENTREPRENEURSHIP

Proyecto Vigía — Spain

Co-founder (50%) & sole developer

2026 – Present

- Designed and built the entire platform end to end, single-handedly: Java / Spring Boot backend, React / TypeScript frontend, identity and access management, geospatial PostgreSQL, document processing services, GDPR machinery and compliant e-invoicing — from architecture to production deployment.
- B2B marketplace for a regulated Spanish services sector, currently in closed beta. Own monetisation strategy, fraud-risk design, infrastructure cost optimisation and the partnership protection framework — the commercial and architectural calls a founder has to make before there is a team to delegate to.

LANGUAGES

Spanish — native. English — full professional proficiency. French — full professional proficiency, with an active francophone network built over four years in France.